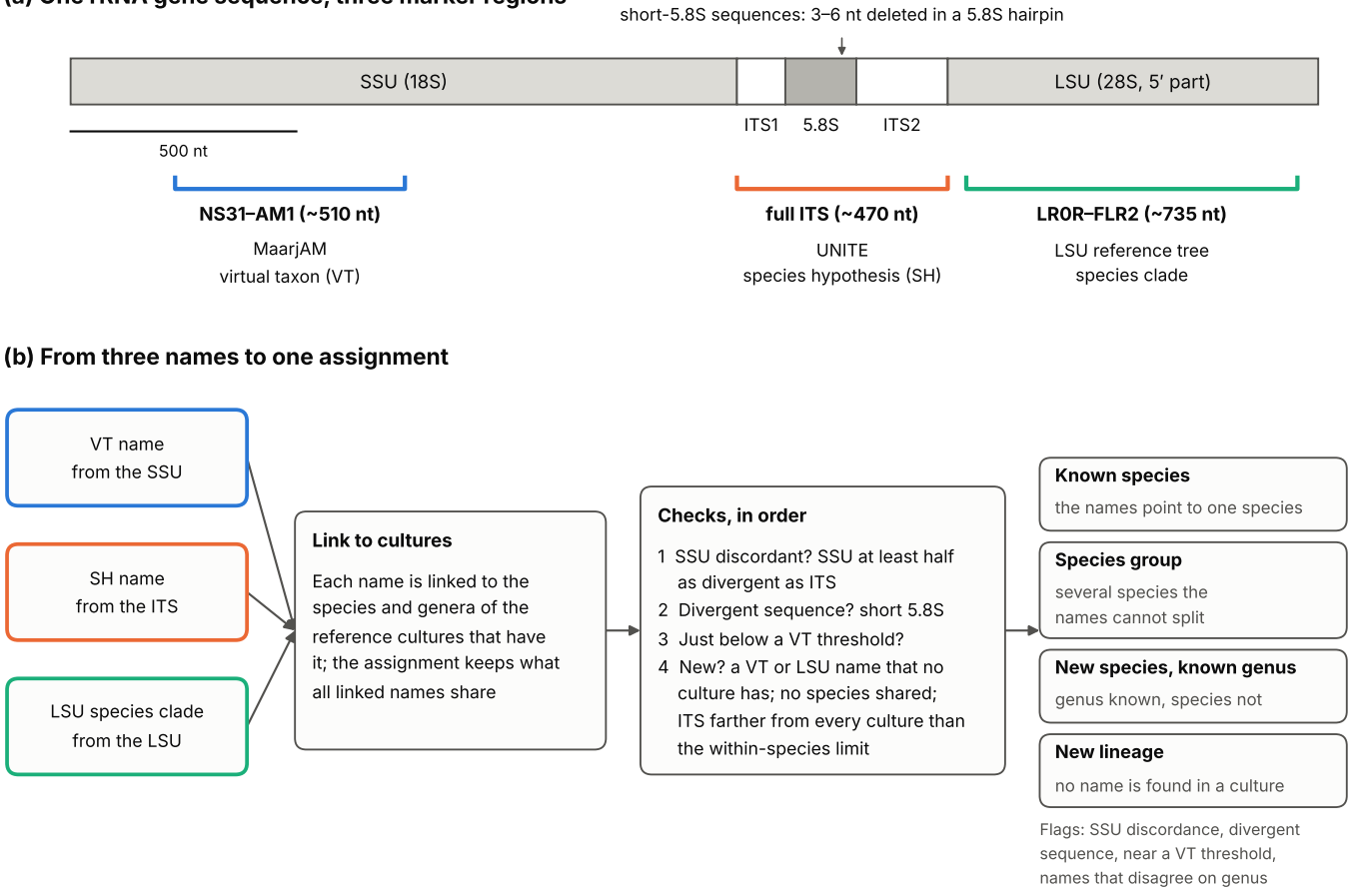
(a) One rRNA gene sequence, three marker regions
short-5.8S sequences: 3–6 nt deleted in a 5.8S hairpin
SSU (18S)
ITS1
5.8S
ITS2
LSU (28S, 5′ part)
500 nt
NS31–AM1 (~510 nt)
MaarjAM
virtual taxon (VT)
full ITS (~470 nt)
UNITE
species hypothesis (SH)
LR0R–FLR2 (~735 nt)
LSU reference tree
species clade
(b) From three names to one assignment
VT name
from the SSU
SH name
from the ITS
LSU species clade
from the LSU
Link to cultures
Each name is linked to the species and genera of the reference cultures that have it; the assignment keeps what all linked names share
Checks, in order
1 SSU discordant? SSU at least half as divergent as ITS
2 Divergent sequence? short 5.8S
3 Just below a VT threshold?
4 New? a VT or LSU name that no culture has; no species shared; ITS farther from every culture than the within-species limit
Known species
the names point to one species
Species group
several species the names cannot split
New species, known genus
genus known, species not
New lineage
no name is found in a culture
Flags: SSU discordance, divergent sequence, near a VT threshold, names that disagree on genus